

Experiment No.5

Aim: To study and implement Platform as a Service using AWS Elastic Beanstalk/Microsoft Azure App Services.

Objectives:

- Develop a theoretical understanding of Platform as a Service (PaaS) and its significance in cloud computing
- Apply practical knowledge to deploy and manage applications on AWS Elastic Beanstalk.

Outcomes:

- Students will demonstrate proficiency in utilizing AWS Elastic Beanstalk to deploy, scale, and manage web applications.
- Students will gain applied knowledge in deploying and managing applications on a PaaS environment.

Software Requirements: Windows/Ubuntu OS, AWS Account.

Theory:

Platform as a Service (PaaS) is a cloud computing category that provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the underlying infrastructure. AWS Elastic Beanstalk is an AWS service that simplifies the deployment of applications by automatically handling the capacity provisioning, load balancing, and application health monitoring. It supports multiple programming languages and allows developers to focus on writing code while AWS manages the platform.

In Elastic Beanstalk, applications are deployed in an environment that consists of Amazon EC2 instances, an Auto Scaling group, and a load balancer. Elastic Beanstalk can automatically scale the number of instances based on application demand, ensuring optimal performance and resource utilization.

Key Components:

1. **Application Code:** Your web application's source code and assets, ready for deployment
2. **Elastic Beanstalk Service:** Manage the deployment, configuration, and scaling of your application across multiple AWS resources.
3. **EC2 Instances:** Virtual machines that host your application code and serve user requests.
4. **Load Balancer:** Distributes incoming traffic across multiple EC2 instances, ensuring high availability and performance.
5. **Auto Scaling Group:** Automatically scales the number of EC2 instances based on your application's workload, optimizing resource utilization.
6. **Other AWS Services:** Elastic Beanstalk seamlessly integrates with other AWS services like S3 for storage, RDS for databases, and CloudWatch for monitoring.

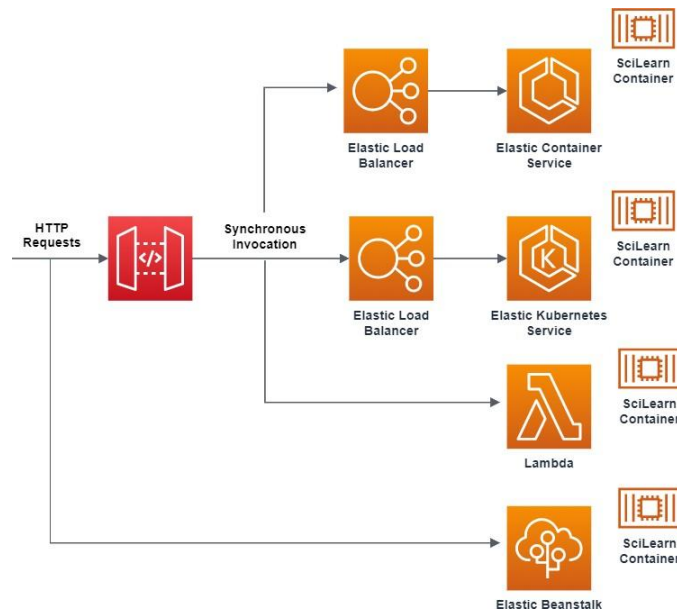


Figure 6 Elastic Beanstalk's Architecture

How It Works:

1. **Upload Code:** You provide your application code and configuration details to ElasticBeanstalk.
2. **Environment Creation:** Elastic Beanstalk provisions the necessary AWSresources, including EC2 instances, load balancers, and Auto Scaling groups.
3. **Deployment:** Your application is deployed to the provisioned environment and madeaccessible to users.
4. **Health Monitoring:** Elastic Beanstalk continuously monitors your application's health and performance.
5. **Auto Scaling:** If your application experiences increased traffic, Elastic Beanstalkautomatically scales up the resources to handle the load.
6. **Management Simplification:** You can manage your application environment through the Elastic Beanstalk console or APIs.

Benefits of Elastic Beanstalk:

- **Simplified Deployment:** Streamlines application deployment with minimalconfiguration.
- **Automatic Scaling:** Ensures optimal resource utilization and performance.
- **Managed Infrastructure:** Eliminates the need to manage individual AWS resourcesmanually.
- **Cost-Effectiveness:** Pay only for the resources you use, eliminating upfront infrastructure costs.
- **Enhanced Security:** Benefits from AWS's robust security measures.
- **Integration with Other AWS Services:** Works seamlessly with a wide range of AWSservices.

Stepwise Procedure to Implement PaaS using AWS Elastic Beanstalk:

Step 1: Sign into AWS Console

- Log in to the AWS Management Console using your AWS account credentials.

Step 2: Access Elastic Beanstalk Service

- Navigate to the Elastic Beanstalk service in the AWS Console.

Step 3: Create a New Application

- Click on "Create Application."
- Provide a name for the application and choose a platform (e.g., Node.js, Python, Java).
- Configure additional settings, such as environment type and platform branch.

Step 4: Upload Application Code

- Upload your application code or provide a link to the application source code repository.
- Configure application settings, such as environment variables.

Step 5: Configure Advanced Settings

- Customize advanced settings, including instance types, scaling options, and security settings.
- Review and modify other configuration options as needed.

Step 6: Launch Environment

- Click on "Create environment" to launch the Elastic Beanstalk environment.
- The environment will include Amazon EC2 instances, an Auto Scaling group, and a load balancer.

Step 7: Monitor Deployment

- Monitor the deployment process in the Elastic Beanstalk console.
- Verify that the environment is successfully created, and the application is deployed.

Services
Resource Groups

ternadgtester1 @ 5537-756...
N. Virginia
Support

AWS Management Console

AWS services

Find Services
You can enter names, keywords or acronyms.

ElastiCache
In-Memory Cache
Elastic Beanstalk
Run and Manage Web Apps
Elastic Transcoding
Easy-to-Use Scalable Media Transcoding
Elasticsearch Service
Run and Scale Elasticsearch Clusters
EC2
Virtual Servers in the Cloud
ECR
Fully-managed Docker container registry
EFS
Managed File Storage for EC2
EMR
Batch
Elastic Beanstalk
Rekognition
Amazon Transcribe

Access resources on the go

Access the Management Console using the AWS Console Mobile App. [Learn more](#)

Explore AWS

AWS Marketplace

Find, buy, and deploy popular software products that run on AWS. [Learn more](#)

Amazon Redshift

Fast, simple, cost-effective data warehouse that can extend queries to your data lake. [Learn more](#)

Run Serverless Containers with AWS

Elastic Beanstalk

[Create New Application](#)

Welcome to AWS Elastic Beanstalk

With Elastic Beanstalk, you can **deploy**, **monitor**, and **scale** an application quickly and easily. Let us do the heavy lifting so you can focus on your business.

To deploy your **existing web application**, create an application source bundle and then create a new application. If you're using Git and would prefer to use it with our command line tool, please see [Getting Started with the EB CLI](#).

To deploy a **sample application**, click **Get started**, choose a name, select a platform and click **Create app**.

By launching the sample application, you allow AWS Elastic Beanstalk to administer AWS resources and necessary permissions on your behalf. [Learn more](#)

[Get started](#)

Get Started in Three Easy Steps

Select a Platform

Upload an Application or Use a Sample

Run it!

Services
Search

[Alt+S]
N. Virginia
voclabs/user2363114=Soheil @ 5404-0088

Elastic Beanstalk

Environments
Applications
Change history

Elastic Beanstalk
Getting started

Create a web app

Create a new application and environment with a sample application or your own code. By creating an environment, you allow Amazon Elastic Beanstalk to manage Amazon Web Services resources and permissions on your behalf. [Learn more](#)

Application information

Application name

Up to 100 Unicode characters, not including forward slash (/).

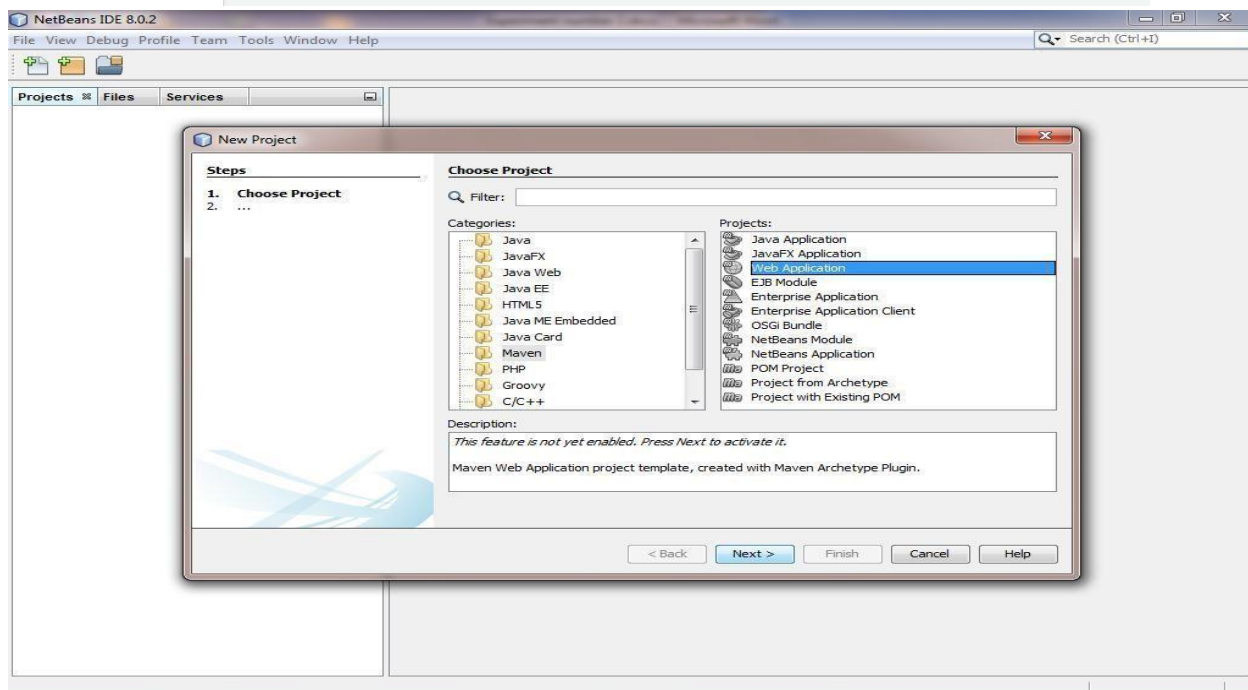
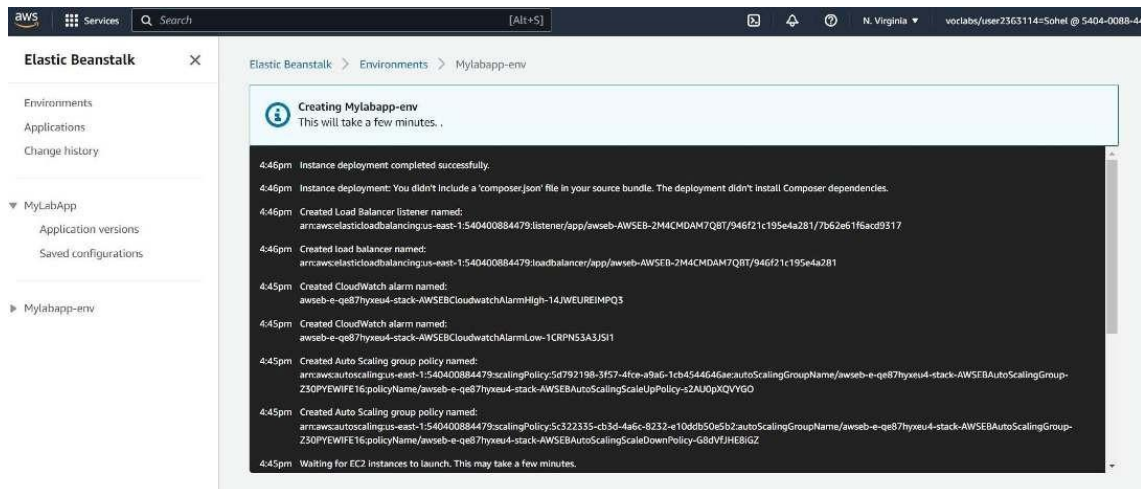
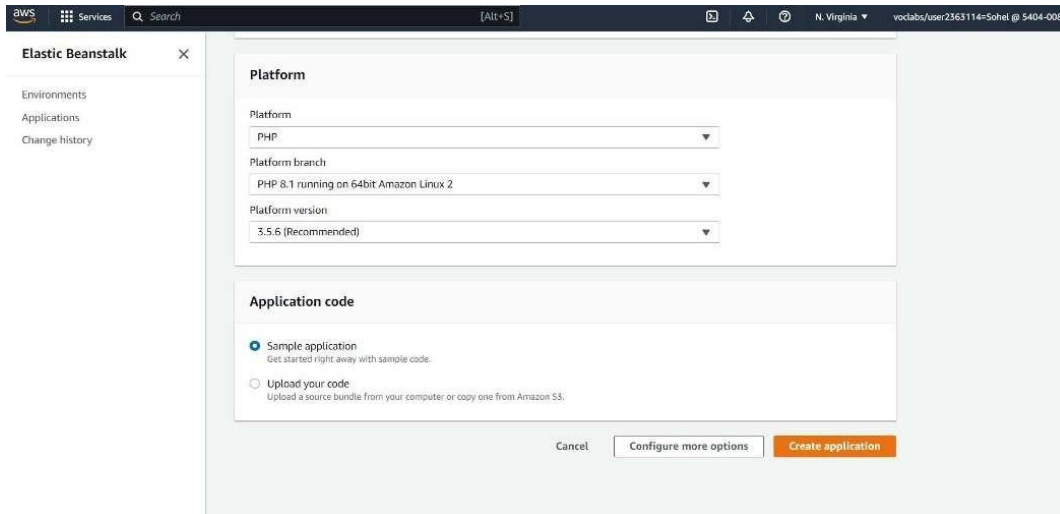
Application tags

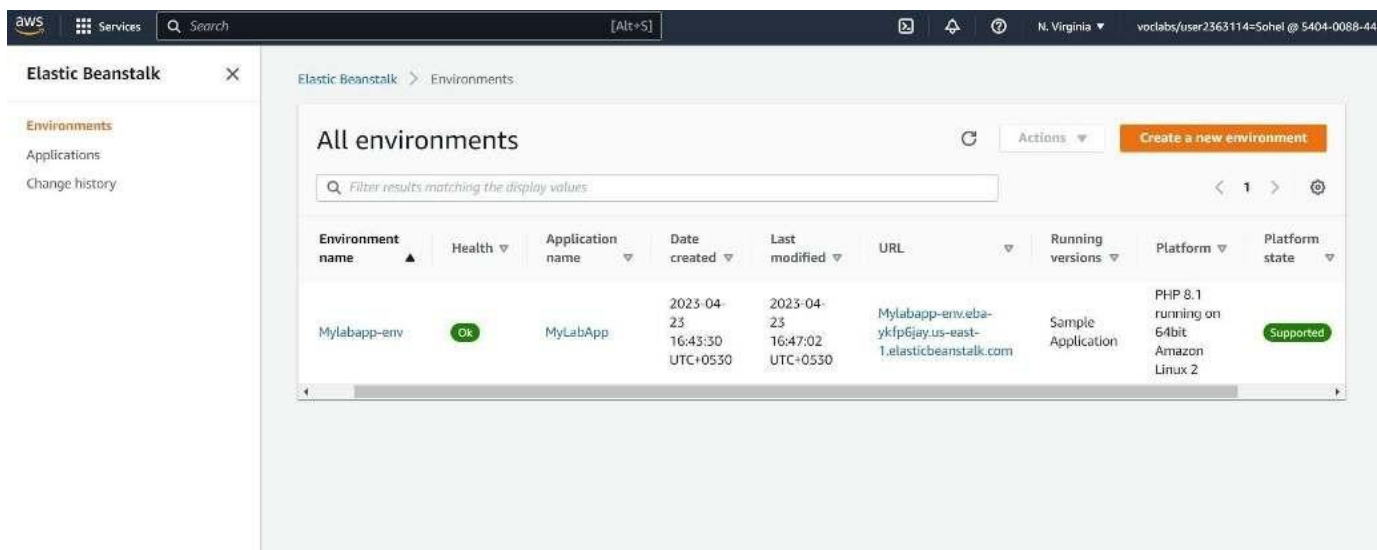
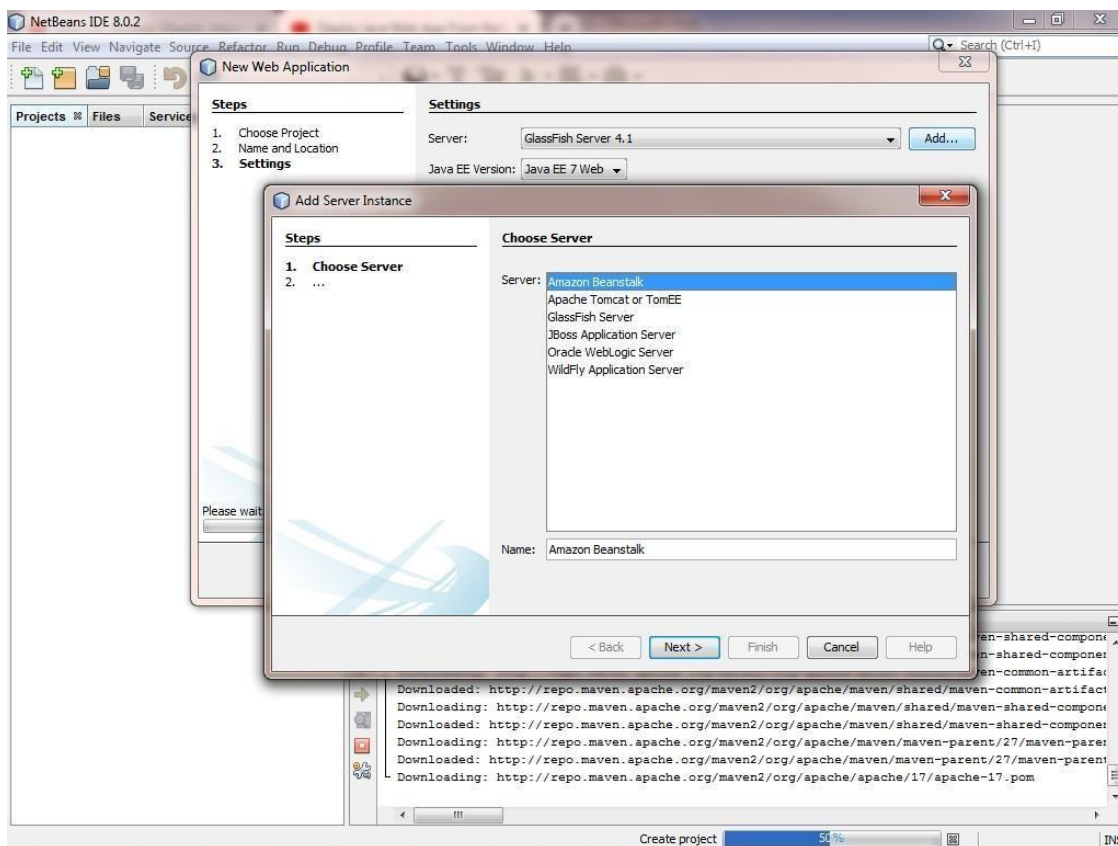
Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. [Learn more](#)

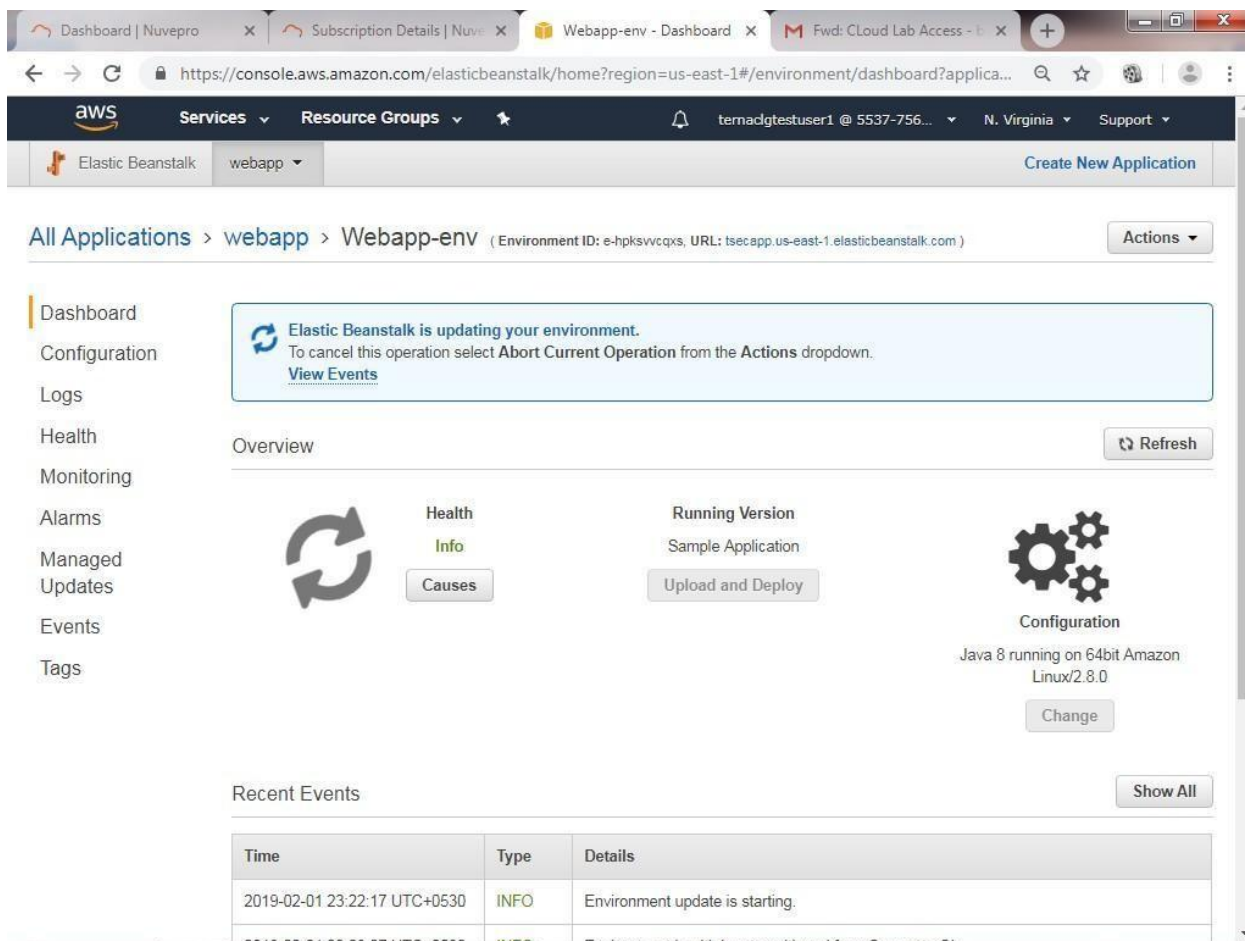
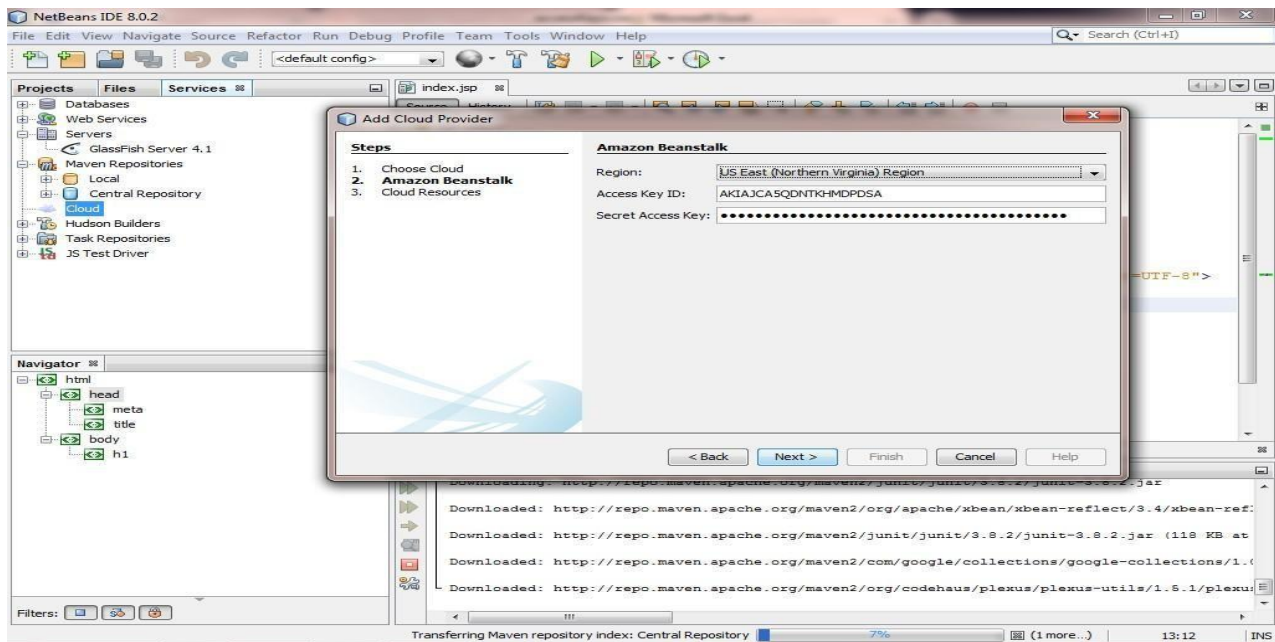
Key	Value	
		Remove tag

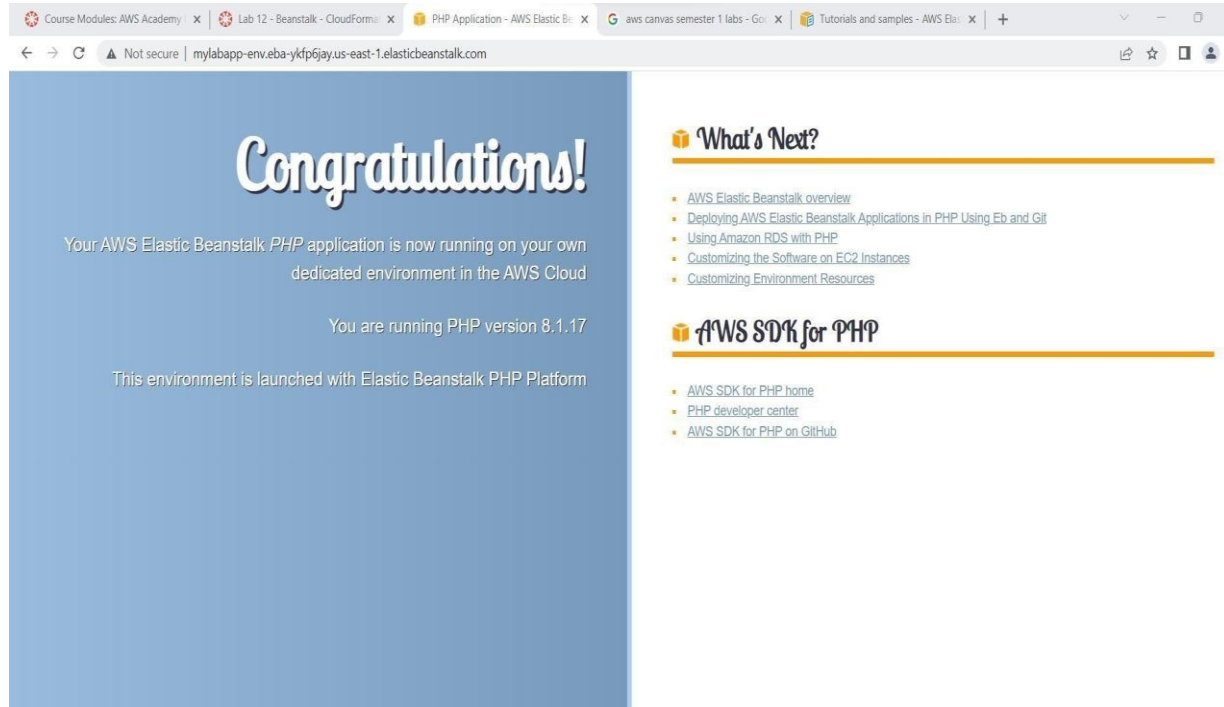
[Add tag](#)

50 remaining









The application deploys to the environment using all of the cloud resources Elastic Beanstalk provisioned. To see your PHP website, in the left navigation pane, choose **Go to environment**. The web application opens in a new tab.

Conclusion:

Thus, we have hands-on experience in implementing Platform as a Service (PaaS) using AWS Elastic Beanstalk.